ANOVA & Sum of Squares v3

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Based off of 9ST0 content.
lbweb.uk/publications

Analysis of Variance, or ANOVA, is to test means across three or more groups to see if at least one of them is different to the rest of the samples.
This is done by looking at the Spread (Variance) between the Samples rather than within the Samples.
If the variation between groups is massively larger than the random noise inside them, you conclude the means are not all equal.

Blocking factors are when you intentionally group data together to isolate and remove variation caused by external factors.

- e.g. You have a sample of 12 plants however 5 used a seperate and less expensive fertiliser. This is therefore likely to cause variance in the sample, so we group each plant by the fertiliser used. Another blocking factor in the same sample, blocking factor B, may be the location they grew in a warehouse.
- Adding these blocking factors mean you get a smaller residual/error

Residual / Error is the representation of unaccounted for unpredictable real-world variation that your model cannot explain. It's whatever is left after we calculate how much variation is caused by either blocking factor

Sum of Squares is the measure of the total amount of variation in your data.

Degrees of freedom is the representation of how many independant values are able to vary once the mean is calculated.

- e.g. if you have a sample size of 5 with mean 10, $df = 5-1 = 4$.
  Therefore 4 values are able to vary from the mean thus creating variation, and that 1 value must stay the same to ensure the mean stays at 10.

Mean of Squares is the average varience per degree of freedom.

F-number is your test statistic.

Required Formulae:

- Sum of Squares (SS) → $SS_T = \sum x_{ij}^2 - \frac{(\sum x)^2}{n}$
  - Where x_{ij} represents $x_{row,\ column}$
    - e.g. $x_{2,3}$ represents the value in row 2 column 3
- Sum of Squares for a blocking factor → $SS_A = \sum \frac{T_i}{n_i} - \frac{(\sum x)^2}{n}$
  - Where n_i represents the number of values in a row or column
  - Where T_i is the row or column's total
- Residual / Error Sum of Squares → $SS_{R/E} = SS_T - (SS_A + SS_B)$
- Degrees of freedom (df) → $df = n_i - 1$
  - Where n_i is either the number of values in a row or column, or total number of values.
- Mean of Squares (MS) → $MS = SS \div df$
- F number → $F = MS \div MS_R$
  - Where MS_R is the mean of Squares of the residual / error band.
- As shown by $SS_{R/E}$, $MS_{R/E}$ and $df_{R/E}$ is calculated the same way just with df or ms instead of ss

Example

Example Data		Camper		
		Olan	Majid	Ralphet
Model	A	42	29	19
	B	37	33	24
	C	24	29	18
	D	25	22	13

An outdoor equipment researcher investigates whether the sleep score achieved by campers depends on the tent model used. Four tent models (A, B, C, and D) are tested. Three campers (Olan, Majid, and Ralphet) act as blocking factors, with each camper testing all four tent models in randomized trials.

The resulting sleep scores (out of 50) are recorded in the table

Statistics Mode on the CG50 where data is all under List 1 and frequency is 1

$\Sigma x = 315$

$\Sigma x^2 = 9019$

$n = 12$

To calculate the "ANOVA Table", either use Stats mode or formulae below:

$$SS_{Total} = 9019 - \frac{315^2}{12} = 750.25$$

Example Data		Camper			T
		Olan	Majid	Ralphet	
Model	A	42	29	19	90
	B	37	33	24	94
	C	24	29	18	71
	D	25	22	13	60
	T	128	113	74	315

Camper is blocking factor A

Model is blocking factor B

Remember, $SS_A = \Sigma \frac{T_i}{n_i} - \frac{(\Sigma x)^2}{n}$

$$\therefore SS_A = \left(\frac{128^2}{4} + \frac{113^2}{4} + \frac{74^2}{4}\right) - \frac{315^2}{12} = 388.5$$

→ Since each Camper has 4 values

$$\therefore SS_B = \left(\frac{90^2}{3} + \frac{94^2}{3} + \frac{71^2}{3} + \frac{60^2}{3}\right) - \frac{315^2}{12} = 256.92$$

→ Since each Model has 3 values

Remember, $SS_{R/E} = SS_T - (SS_A + SS_B)$

$\therefore SS_{R/E} = 750.25 - (388.5 + 256.92) = 104.83$

Remember, $df = n - 1$

$\therefore df_A = 3 - 1 = 2$ → Since 3 Campers

$\therefore df_B = 4 - 1 = 3$ → Since 4 Models

$\therefore df_T = 12 - 1 = 11$

$\therefore df_{R/E} = 11 - (3 + 2) = 6$

Remember, $MS = SS \div df$

$\therefore MS_A = \frac{388.5}{2} = 194.25$

$\therefore MS_B = \frac{256.92}{3} = 85.64$

$\therefore MS_{R/E} = \frac{104.83}{6} = 17.47$

Remember, $F = MS \div MS_{R/E}$

$\therefore F_A = \frac{194.25}{17.47} = 11.12$

$\therefore F_B = \frac{85.64}{17.47} = 4.902$

ANOVA Test, One-tailed at 5% Significance.

Factor A:

> H_0: $\mu_O = \mu_M = \mu_R$

>> where μ_O represents the population mean of Sleep Scores for Camper Olan

>>> μ_M represents the population mean of Sleep Scores for Camper Majid

>>> μ_R represents ...

> H_1: at least two $\mu_i \neq \mu_j$

Factor B:

> H_0: $\mu_A = \mu_B = \mu_C = \mu_D$

>> where μ_A represents the population mean of Sleep Scores for tent Model A.

>> where μ_B, μ_C, μ_D ...

> H_1: at least two $\mu_i \neq \mu_j$

Too little space to write all fully. Sorry!

Finally giving us the following:

Results/Answers	SS	DF	MS	F
BF. A	388.5	2	194.25	11.12
BF. B	256.92	3	85.64	4.902
Error	104.83	6	17.47	—
Total	750.25	11	—	—

Comparing against the CV:

Databooklet Page 21 & 22. Use the table with your α
Here, we use the table where "$(p = 0.95)$" since our $\alpha = 1 - 0.05 = 0.95$
V_1 is your blocking factors df and V_2 is your error's df
Then compare both blocking factors to their respective CV. (F value)

CV_A: $V_1 = 2$ & $V_2 = 6$ $\therefore CV_A = 5.143$ $\therefore 11.12 > 5.143$ $\therefore$ Fail to accept H_0
CV_B: $V_1 = 3$ & $V_2 = 6$ $\therefore CV_B = 4.757$ $\therefore 4.902 > 4.757$ $\therefore$ Fail to accept H_0
→ insert contextual analysis here.

If using the CG50:

List 1 is your data going like this:

Example Data		Camper		
		Olan	Maiid	Ralphet
Model	A	42	29	19
	B	37	33	24
	C	24	29	18
	D	25	22	13

Deg Norm1 d/c Real

SUB	List 1	List 2	List 3	List 4
1	42	1 (Olan)	1 (Model A)	1
2	37	1	1 (Model B)	2
3	24	1	1 (Model C)	3
4	25	1	1 (Model D)	4
5	29	2 (Maiid)	2 ↓etc.	1
6	33	2	2	2
7	29	2	2	3
8	22	2	2	4
9	19	3 (Raphlet)	3	1
10	24	3	3	2
11	18	3	3	3
12	13	3	3	4

List 2 is the position the values in within blocking factor A

List 3 is the position the values in within blocking factor B.

GRAPH CALC TEST INTR DIST ▷
F3 → F5
Z t CHI F ANOVA

If you have only one blocking factor, Set "how many?" to 1 and leave list 3 (blocking factor B) blank.

Deg Norm1 d/c Real
ANOVA
How Many:2
Factor A:List2
Factor B:List3
Dependnt:List1
Save Res:None
GphColor:Magenta
1 2

Deg Norm1 d/c Real
ANOVA

	df	ss	ms	F →	P
A	2	388.5	194.25	11.117	9.5E-3
B	3	256.91	85.638	4.9014	0.047
ERR	6	104.83	17.472		

p is the probability of the F-value occurring if we accept H_0 ← Same as Chi-Squared

ERR is error or residual.
A and B are the blocking factors.